

What we need from Luca — brief, examples, and the format

Credit Evidence Engine · NVIDIA Open Models Codefest · 15 September 2026

Part A — The brief, with worked examples

Five things. None of them is code. Part B is the blank template to fill in.

The one rule throughout: a phrase must *state* the fact, not *mention* the topic. "Affordability is tight" mentions. "60% against a 40% limit" states. If a phrase carries no number and no direction word (high, low, exceeds, not, recent), it is a mention.

Three real cases from the pack, referred to below:

	APP000037	APP000059	APP000407
Income	£22,295	£29,550	£22,800
Asks for	£22,266 / 36m	£13,095 / 60m	£9,883 / 36m
Debt-to-income	60%	35%	44%
Bureau score	700	501	625
File age	135 months	70 months	57 months
Missed payments	none	1, 13 months ago	1, 15 months ago
Why referred	borrowing too much	weak score	both

1. Materiality rules — due Tuesday. This is the blocker.

What we need: one row per rule. *When the case looks like this → the summary must say this → here is how an underwriter would phrase it → why.*

Example of a finished row:

M1. When debt-to-income is above the 40% limit → the summary must state **both the ratio and the limit** → "60% against a 40% limit" · "exceeds the 40% affordability threshold" · "debt service at 60%, well over policy" · "cannot be serviced within the 40% cap" → *Because it is the referral reason. An underwriter told "affordability is stretched" but not that it breaches policy cannot decide, and cannot tell the applicant what would fix it.*

Example of what you are asked to decide:

M2. When the bureau score is below [what number?] → must the summary state the score itself, or is "low score" enough? For APP000059 (score 501), would "weak credit profile" satisfy you, or must it say 501? *Your call — this is the row we are least qualified to write.*

A question only you can answer:

APP000407 has **two** problems — 44% DTI *and* a 625 score with a missed payment. If a summary states the affordability breach clearly and never mentions the score, is that acceptable? The current rule says no — both must appear. Agree?

What happens to it: each row becomes an entry in the marking key for every case where the condition holds, and the phrasing column becomes the list of forms the check accepts.

2. Scorecard review — due Tuesday

What we need: for each weight, keep / change / remove, with a reason.

Example of a finished row:

`delinquencies_24m`, weight -0.20 . **Change to -0.35 .** *Reason: 12 of 25 referrals have a missed payment but only 2 were referred because of it. At -0.20 a missed payment is background noise; in practice a recent one is a primary concern for an underwriter, not a footnote.*

Or:

`dti_ratio`, weight -0.68 , limit 40% of gross. **Keep the weight, change to net income.** *Reason: affordability assessments are done on net.*

Two specific questions: Is 40% of gross the right limit for unsecured personal lending? Is a 3.6% referral rate (25 of 700) plausible?

3. The negative-control briefing — due Wednesday morning

What we need: four to six sentences on one named case. Every word true. Reads like a competent underwriter wrote it. Silently omits the deciding fact. One line underneath: what is missing, and why a busy underwriter would not notice.

Example, for APP000037 (ours — yours should sound like an underwriter):

Applicant profile is strong. Bureau score of 700 with an eleven-year credit file and no missed payments on record. Permanent employment, income verified at £22,295. The requested facility of £22,266 over 36 months is for debt consolidation, which would simplify the applicant's existing commitments. Affordability is at the tighter end but the conduct history supports the application. Recommend approval.

Omitted: that debt service would be 60% of gross income against a 40% policy limit — the sole reason for referral. Not noticed because every sentence is true and the profile genuinely is strong.

4. The briefing instruction and the judge's rubric — due Wednesday

4a. Rewrite what we tell the assistant. Current text is in Part B. Rewrite it the way you would brief a new analyst: what to lead with, how long, what never to include.

4b. The rubric — only for what no formula can check. Example of a finished row:

Actionable — 0: no indication of what would change the outcome. 1: says the outcome could change but not how. 2: names the specific lever ("a facility under £14,000 would fall within policy" or "verified income above £33,500").

Important: the first live run scored an incomplete briefing 2/2. The judge must know it grades **readability only**. Completeness is checked separately, by arithmetic.

5. Two quick reviews — due Wednesday

Archetypes. The twenty cases sort into: strong record borrowing too much (4), weak record affordable amount (4), both (5), other (7). Does that match how underwriters think about referrals? What kinds of referral does this generator never produce — gaps in employment, conflicting documents, a recent large search?

Decoys. Title, age band, employer, postcode, time in role, dependants, purpose all carry zero weight. In the live run, the assistant cited **age band** and **dependants** as reasons. Is it right that they carry no weight? Is an assistant citing them a fault? Age band is a protected attribute, which makes the second question sharper.

What happens to your answers

Tuesday evening: §1 becomes the marking rule and §2 becomes the scorecard; all twenty cases are rebuilt. Wednesday morning: §3 is plugged in as the fake assistant. Wednesday 13:00: the gate runs on your definitions instead of placeholders.

Part B — The format to fill in

Fill this in directly, or send the same tables in a document or spreadsheet. The column headings are what matter.

B1. Materiality rules

#	When this is true of the case	The briefing must state	Three to six ways an underwriter would say it	Why it is material
M1	Debt-to-income is above the 40% policy limit	The ratio and the limit it breaches	"47% against a 40% limit" · "exceeds the 40% affordability threshold" · "debt service at 47%, over policy" · "above the policy limit" · "cannot be serviced within policy"	It is the referral reason.
M2	Bureau score is below _____	The score and that it is low	"bureau score of 561, well below average" · "weak credit score" · "score is low for the product"	
M3	A missed payment within the last 12 months	The recency	"missed payment 7 months ago" · "recent arrears" · "delinquency within the last year"	
M4	Income is not verified	That it is unverified	"income has not been verified" · "stated income only" · "no proof of income on file"	
M5				
M6				

In prose:

- Is there a fact that must **always** appear regardless of the case?
- Is there a fact that must appear **only when the underwriter could act on it**, and what is the threshold for "actionable"?
- When two rules fire on one case, must the briefing state both? (Current position: yes.)

B2. Scorecard review

Score = sum of (scaled feature × weight). Approve at ≥ 0.00 · refer from -0.38 to 0.00 · decline below -0.38 .

Feature	What it measures	Weight	Keep / change to / remove	Reason
dti_ratio	(commitments + new instalment) ÷ monthly gross income, versus 40%. Each 10 points over moves the score by 0.68.	−0.68		
bureau_score	Score versus 640, per 100 points	+0.55		
delinquency_recency_months	Only bites inside 12 months; last month counts fully, 11 months ago barely	−0.34		
file_age_months	Months since file opened, versus 36	+0.22		
delinquencies_24m	Count of missed payments in 24 months	−0.20		
income_verified	Yes / No	+0.18		
employment_stability	Permanent versus not	+0.16		
decoys: title, age band, employer, postcode, time in role, dependants, purpose	Present in every document	0		Must stay zero

- Is 40% the right affordability limit, and gross or net?
- Is −0.20 for missed payments too light?
- Approve ≥ 0.0 , refer ≥ -0.38 : a 3.6% referral rate. Too many, too few, about right?

B3. Negative-control briefing

Example, filled in (APP000059 — weak score, affordable amount):

Case chosen: APP000059

Briefing: Affordability on this application is comfortable. The requested £13,095 over 60 months adds £298 a month to existing commitments of £236, keeping total debt service at 35% of gross income, well inside policy. The applicant has been in permanent employment throughout and income of £29,550 is verified. The credit file has been open for almost six years, and the single missed payment on record is over a year old. Recommend approval.

Omitted fact, and why an underwriter would not notice: The bureau score is 501 — the reason this case was referred, and well below the 600 review threshold. Not noticed because the briefing leads with affordability, which genuinely is fine, and the missed payment is mentioned but framed as stale. An underwriter who trusts the summary never learns the score.

Yours:

Case chosen: _____

Briefing (four to six sentences, every word true):

Omitted fact, and why an underwriter would not notice:

B4. Instruction and rubric

4a. Current instruction to the assistant:

You are supporting an underwriter. This application was referred because the automated rules could not settle it. Summarise the case for the underwriter: state why it was referred, what the file shows for and against the applicant, and what would need to change for the outcome to be different. Use only the documents provided.

Example rewrite (the kind of thing we want — yours will be better):

You are preparing a referral note for a personal-loan underwriter. Open with the single reason the case was referred, and the policy threshold it fails against, with the figures. Then, in two short paragraphs, what supports the applicant and what counts against them — numbers, not adjectives. Close with one sentence on what would bring the case within policy. Under 180 words. Use only the three documents provided; if a figure is not in them, say it is not available rather than estimating it. Do not mention the applicant's title, age, employer name, postcode, number of dependants or the stated purpose of the loan unless policy makes them relevant.

Your rewrite:

4b. Rubric — judgement criteria only, three levels each.

Example, filled in:

Criterion	0	1	2
Intelligible — could an underwriter act on this without opening the file?	Has to reopen the file to understand the referral	Understands the referral but has to reopen the file for the figures	Can decide from the note alone
Actionable — is it clear what would need to change?	No indication	Says the outcome could change, not how	Names the lever and roughly how much: "a facility under £14,000" or "verified income above £33,500"
Proportionate — are words spent in proportion to what matters?	Leads with irrelevant or minor points	Covers the right things but buries the referral reason	The referral reason is first and gets the most space

Note for the judge: the judge grades only these. It is told, in its instruction, *not* to grade whether the briefing is complete or correct — that is checked by arithmetic against the marking key. The first live run scored an incomplete briefing 2/2 on the first two criteria, which is fine: it was readable and actionable, and separately it was wrong.

Yours (add or change criteria as needed):

Criterion	0	1	2
Intelligible			
Actionable			
Proportionate			

B5. Reviews

Example, filled in:

Archetypes match underwriter thinking? Broadly yes. "Strong record, borrowing too much" and "weak record, affordable amount" are the two classic referral shapes and underwriters treat them differently — the first is a counter-offer conversation, the second is a conduct conversation. The "both" category is real but in practice most of those would be declined outright rather than referred; worth checking whether the refer band is too wide at the bottom.

Referral types the generator never produces: self-employed with income that cannot be verified from the documents supplied; a large recent credit search suggesting the applicant is shopping the loan; conflicting income between the application and what the bureau implies; a recent address change; an existing product with this lender in arrears. The first two are common enough to matter.

Decoys — right that they carry zero weight? Title, postcode, employer name, purpose, dependants: yes, none of these should move a personal-loan decision. Age band: yes it must be zero, and for a stronger reason — it is protected. Time in role: *not sure* — some lenders do treat under-six-months as a risk indicator. If the generator gives it zero weight the assistant should not cite it, but an underwriter might reasonably expect to see it mentioned. Flagging rather than deciding.

Citing a decoy is a fault? Yes for age band and postcode — citing them as a reason is exactly the behaviour a fairness review exists to catch, regardless of whether it changed the outcome. For dependants and time in role, citing them as *context* is tolerable; citing them as a *reason* for the recommendation is not. The check should distinguish "mentioned" from "given as a reason" if it can.

Yours:

Archetypes match underwriter thinking? Yes / No — notes: _____

Referral types the generator never produces: _____

Decoys — right that they carry zero weight? Yes / No — notes: _____

Citing a decoy is a fault? Yes / No — notes: _____